

containing all the elements of the group will be initiated by each of them.

IX. CONCLUSIONS

An algebraic method for determining the irredundant prime closed sets (IPCS) has been discussed in this note. The elegance of the method lies in the fact that all the IPCSs initiated by different classes are obtained at a time through certain algebraic operations. Once these IPCSs corresponding to a flow table are obtained, the minimal cover of the flow table can be readily determined with the help of a cover chart [1].

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On Simplifying Truth Functions: A Preliminary Reduction of Coreless Formulas

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Abstract—The core of a truth function ϕ consists of those prime implicants of ϕ which are terms of every disjunctive normal formula equivalent to ϕ . The notion of core is generalized here: a class of prime implicants of ϕ is defined, called its "pseudo core." The pseudo core of ϕ consists of those prime implicants of ϕ which are terms of every one of a class Γ of normal equivalents of ϕ , where Γ includes a simplest normal equivalent. The pseudo core may be found easily and rapidly, plays the same role as the core in simplifying a formula, and may exist even in coreless formulas.

Index Terms—Core, prime implicant, simplest normal equivalent, truth function.

INTRODUCTION

A coreless formula ϕ may be simplified by surveying all possible sums of its prime implicants, and selecting one which is irredundant, equivalent to ϕ , and has the least

number of terms. Many variants and refinements of this procedure are known, incorporating useful simplifications and short cuts, but none of them is able to avoid the basically exhaustive character of the above routine. For this reason, if a formula has a core, the preliminary determination of the core, and of the prime implicants which are covered¹ by other prime implicants, can save a great deal of the labor of simplification. It is therefore natural to ask if there is not a way to carry this preliminary reduction a step further.

The following ideas motivated the research which is reported in this note. The core of a formula ϕ is that sum of prime implicants of ϕ which figures in every normal equivalent of ϕ , in particular, in every simplest normal equivalent of ϕ . However, inasmuch as our aim is to find just one simplest normal equivalent of ϕ , it appears that the core of ϕ satisfies more restrictive conditions than are really needed. Thus, if it were possible to identify prime implicants of ϕ whose sum is part of at least one simplest normal equivalent of ϕ , then this sum could be used in the same way as the core; furthermore, it could obviously be made to include the core, and may exist even where ϕ has no core.

The purpose of this note is to show that such results can indeed be found, and that they may constitute an interesting new line of investigation. The principal result of this note is a criterion that identifies prime implicants of a formula ϕ which do not belong to its core but which are terms of every one of a class Γ of normal equivalents of ϕ , where Γ includes a simplest normal equivalent. This criterion also determines prime implicants of ϕ which are not terms of any formula belonging to the class Γ . The value of knowing these two classes of prime implicants is obvious: ϕ has a simplest normal equivalent which includes all the prime implicants in the first class and none of those in the second class.

The criterion described herein is such a simple and basic one that most of the coreless formulas which the author encountered in connection with work on switching networks were amenable to reduction by this method.

NOTATION AND TERMINOLOGY

Binary variables and their negations will be referred to collectively as *literals*. Suppose ϕ is a truth-functional formula and that γ is a product of literals in which no literal occurs twice; if $\gamma \Rightarrow \phi$ then γ is called an *implicant* of ϕ . Suppose that γ is an implicant of ϕ which satisfies the following condition: if any single literal is deleted from γ , the resulting product is not an implicant of ϕ ; then γ is called a *prime implicant* of ϕ .

In the remainder of this note we will adhere to the following conventions. We will refer to prime implicants as PIs and to sums of prime implicants as SPIs. We will use lower case Roman letters to denote PIs and capital Roman letters to denote SPIs. If A is the sum $a_1 + a_2 + \cdots + a_n$, then

¹ If a and b are prime implicants of ϕ , K is the core of ϕ , and K is nonempty, we say that a covers b if $b \Rightarrow K + a$. Quine [4] has shown that if b is covered by a prime implicant $a \neq b$, then we can find a simplest normal equivalent of ϕ which does not include b .

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$a_1, a_2, \dots, a_n$ are called the *terms* of A . In the following definitions, let A and B be SPIs and let x and y be PIs.

$x \in A$	will mean that x is a term of A .
$x \notin A$	will mean that x is not a term of A .
$A \subseteq B$	will mean that every term of A is a term of B .
$A \subset B$	will mean that $A \subseteq B$ but not $B \subseteq A$.
$(A - x)$	will refer to the SPI whose terms are all the terms of A with the exception of x (x need not necessarily be a term of A).
$\forall$	will mean "for all."
$\exists$	will mean "there exists."
$\Rightarrow$	will mean "implies."
$\nexists$	will mean "there does not exist."
$\nRightarrow$	will mean "does not imply."

If $\forall x \in A, A \nRightarrow (A - x)$, then we will say that A is *irredundant*; otherwise A is *redundant*. Let P be the sum of all the PIs of ϕ : if $P \nRightarrow (P - x)$, then x is called an *unconditional prime implicant* (UPI) of ϕ . If B is a nonempty sum of UPIs and $y \Rightarrow B + x$, then we will say that x *covers* y .

Quine has shown that it is reasonable to define a *simplest normal equivalent*² of ϕ as an irredundant SPI which, among all irredundant SPIs equivalent to ϕ , has the least number of terms [4]. It is known that the UPIs of ϕ are terms of every simplest normal equivalent of ϕ , and that, if x is a PI which is covered by another PI, then there exists a simplest normal equivalent B of ϕ such that $x \notin B$ [2], [4].

The following two additional concepts will be needed in this note. Let A be an SPI and let x be a PI such that $x \notin A$; if $x \Rightarrow A$ and $\nexists A' \subset A$ such that $x \Rightarrow A'$, then A will be called a *prime consequent* of x . If x and y are PIs, and there is an A such that $x \in A$ and A is a prime consequent of y , then we will say that x *commands* y . We will use the notation $x \text{ cm } y$ to abbreviate " x commands y " and $\neg x \text{ cm } y$ to abbreviate " x does not command y ."

THE PRINCIPAL RESULT

Before coming to our main result we must establish a few lemmas which are very simple in nature. We will omit the proofs where these are immediate.

Lemma 1: If $B \subseteq C$, then $B \Rightarrow C$.

Lemma 2: If $B \Rightarrow A$ and $C \Rightarrow A$, then $B + C \Rightarrow A$.

Lemma 3: If $x \Rightarrow A$ and $A \subseteq B$, then $x \Rightarrow B$.

Lemma 4: If $A \Rightarrow B$, then $A + x \Rightarrow B + x$.

Lemma 5: If $\forall x \in B, x \Rightarrow A$, then $B \Rightarrow A$.

Theorem 1: If $x \Rightarrow A$, then there exists an SPI A^* such that $A^* \subseteq A$ and A^* is a prime consequent of x .

Proof: If $x \Rightarrow A$, then either $\nexists A' \subset A$ such that $x \Rightarrow A'$, or else for some $A'' \subset A$ we have $x \Rightarrow A''$. In the first case we set $A = A^*$ and we are done; in the second case we repeat this argument with A'' taking the place of A . Since A has only finitely many terms, the result is established after a few repetitions of this argument.

Theorem 2: If $\neg a \text{ cm } x$ and $x \Rightarrow A$, then $x \Rightarrow (A - a)$.

Proof: By Theorem 1, there exists a formula $A^* \subseteq A$ which is a prime consequent of x . Now $a \notin A^*$; for if $a \in A^*$ then by definition $a \text{ cm } x$. The conclusion now follows by Lemma 3.

We are now able to state our main result.

Theorem 3: Suppose that K is a nonempty SPI. If $a \notin K$, $a \Rightarrow K$, and a commands no more than one term of K , then ϕ has a simplest normal equivalent B such that $a \notin B$.

Proof: Suppose that $a \notin K$ and $a \Rightarrow K$; we will consider two cases, according as a commands no term of K or a commands exactly one term of K .

Case 1: Suppose that a commands no term of K : we will show that a is not a term of any irredundant normal equivalent of ϕ . To be precise, we will assume that C is an irredundant normal equivalent of ϕ such that $a \in C$, and this will lead to a contradiction. Indeed, $\forall x \in K, x \Rightarrow C$, and $\neg a \text{ cm } x$, therefore by Theorem 2, $x \Rightarrow (C - a)$; it follows by Lemma 5 that $K \Rightarrow (C - a)$. Now $a \Rightarrow K$; therefore $a \Rightarrow (C - a)$ by the transitive property of $\Rightarrow$, and it follows that C is redundant, which is a contradiction.

Case 2: Suppose that $b \in K$, $a \text{ cm } b$, and a commands no term of $(K - b)$. Let C be a simplest normal equivalent of ϕ such that $a \in C$, and let B be the formula $(C - a) + b$. Evidently $a \notin B$ (for $b \in K$ and $a \notin K$, hence $a \neq b$); we will show that B is a simplest normal equivalent of ϕ . Indeed, $\forall x \in (K - b), x \Rightarrow C$, and $\neg a \text{ cm } x$, so by Theorem 2, $x \Rightarrow (C - a)$; consequently by Lemma 5, $(K - b) \Rightarrow (C - a)$. It follows by Lemma 4 that $K \Rightarrow B$, and since $a \Rightarrow K$, we have $a \Rightarrow B$. Now B is $(C - a) + b$, so by Lemma 1, $(C - a) \Rightarrow B$, hence by Lemma 2, $C \Rightarrow B$. Thus we have $\phi \Rightarrow C \Rightarrow B \Rightarrow \phi$, so B is equivalent to ϕ . Finally, to see that B is irredundant and is a simplest normal equivalent of ϕ , we merely recall that B has the same number of terms as C , and that C is a simplest normal equivalent of ϕ .

THE REDUCTION ALGORITHM

Our algorithm is based on Theorem 3; it is in two parts, which are the following.

Rule 1: Let a be a prime implicant of ϕ : if there exists K such that $a \notin K$, K is a prime consequent of a , and a commands no more than one term of K , then we delete a from the list of prime implicants of ϕ . After we have successively deleted all the prime implicants which satisfy this condition, we have a reduced list of PIs of ϕ , and it follows from Theorem 3 that ϕ has a simplest normal equivalent B whose terms all belong to the reduced list.

Rule 2: Let P' be the sum of all the PIs belonging to the reduced list of PIs. Let Γ be the set of all normal equivalents D of ϕ such that $D \subseteq P'$. ϕ may prove to have what we shall call a "pseudo core"; that is, it may have PIs which are terms of every formula belonging to Γ . Indeed, if $P' \nRightarrow (P' - x)$ then x belongs to this pseudo core. The elements of the pseudo core are listed separately, then they are deleted from the list of prime implicants of ϕ .

Our algorithm consists of applying Rule 1, to be followed by Rule 2; after Rule 2 has been applied, it may be possible to apply Rule 1 again, followed once again by Rule 2. We do

² This criterion for "simplest" is due to Quine. Other criteria exist, such as the diode count, the gate and input count, etc.

this repeatedly, until Rules 1 and 2 yield no further result. At this stage, we know that ϕ has a simplest normal equivalent which includes the pseudo core and only prime implicants which remain on our reduced list of PIs.

The author has written a computer program for simplifying truth-functional formulas which incorporates this procedure. The first part of the program calculates the list of PIs and the list of prime consequents of each PI. The prime consequents can be obtained by the method of truth-functional ratios described in Ghazala [1]. However, Quine's algorithm for producing the list of PIs of a formula can be modified so as to produce, simultaneously, the prime consequents of all the PIs. How this information is used can best be demonstrated in an example.

EXAMPLE

In what follows, the italic letters p, q, r, s, t will be used to denote binary variables.

Let ϕ be the formula $pq' + qr' + p'r + p's + ps't$. The list of prime implicants of ϕ is the following (the PIs are labeled $a_1, a_2, \dots, a_{12}$ in order to facilitate subsequent reference to them):

$$\begin{array}{llll} a_1: r's & a_4: rs't & a_7: p'r & a_{10}: pq' \\ a_2: q's & a_5: p's & a_8: pr' & a_{11}: q'r \\ a_3: qs't & a_6: ps't & a_9: p'q & a_{12}: qr' \end{array}$$

The prime consequents are displayed in Table I. The i th row of Table I contains the list of prime consequents of a_i . We also require a table whose i th row contains the list of prime implicants which a_i commands. This table will be called Table II, and can easily be constructed from Table I. (It is readily apparent in Table I that ϕ is a coreless formula, for every prime implicant of ϕ has at least one prime consequent).

Our algorithm consists of examining all the prime consequents in succession. Each one is tested in the following manner. If $a_i + a_j + a_k$ is a prime consequent of a_i , then we inquire (by referring to line i of Table II) whether a_i commands more than one of the three prime implicants a_i, a_j, a_k . If not, then we delete a_i from the list of PIs. If we delete a_i , then we must remove line i from Table I and Table II, and delete every prime consequent in Table I which includes a_i among its terms: for it is clear that we do not want to retain prime consequents with terms that have been deleted from the list of PIs.

In our example, we see that $a_5 + a_8$ is a prime consequent of a_1 , and that a_1 commands a_5 but not a_8 . Therefore we delete a_1 from the list of PIs, remove line 1 from Table I and Table II, and delete from Table I all the prime consequents with a_1 as a term.

Similarly, we see that $a_5 + a_{10}$ is a prime consequent of a_2 , and that a_2 commands a_5 but not a_{10} . Therefore we delete a_2 from the list of PIs. In the same manner we delete a_3 and a_4 from the list of PIs. Table I has now been reduced considerably, and an examination of the reduced table

TABLE I

1.	$a_5 + a_8$	$a_5 + a_{10} + a_{12}$	$a_2 + a_{12}$	$a_2 + a_8 + a_9$
2.	$a_5 + a_{10}$	$a_5 + a_8 + a_{11}$	$a_1 + a_{11}$	$a_1 + a_7 + a_{10}$
3.	$a_6 + a_9$	$a_4 + a_8 + a_9$	$a_4 + a_{12}$	$a_6 + a_7 + a_{12}$
4.	$a_3 + a_{11}$	$a_6 + a_9 + a_{11}$	$a_6 + a_7$	$a_3 + a_7 + a_{10}$
5.	$a_2 + a_9$	$a_2 + a_7 + a_{12}$	$a_1 + a_7$	$a_1 + a_9 + a_{11}$
6.	$a_3 + a_{10}$	$a_4 + a_{10} + a_{12}$	$a_4 + a_8$	$a_3 + a_8 + a_{11}$
7.	$a_9 + a_{11}$			
8.	$a_{10} + a_{12}$			
9.	$a_7 + a_{12}$			
10.	$a_8 + a_{11}$			
11.	$a_7 + a_{10}$			
12.	$a_8 + a_9$			

TABLE II

1.	$a_2,$	a_5				
2.	$a_1,$	a_5				
3.	$a_4,$	a_6				
4.	$a_3,$	a_6				
5.	$a_1,$	a_2				
6.	$a_3,$	a_4				
7.	$a_2,$	$a_3,$	$a_4,$	$a_5,$	$a_9,$	a_{11}
8.	$a_1,$	$a_2,$	$a_3,$	$a_6,$	$a_{10},$	a_{12}
9.	$a_1,$	$a_3,$	$a_4,$	$a_5,$	$a_7,$	a_{12}
10.	$a_1,$	$a_2,$	$a_4,$	$a_6,$	$a_8,$	a_{11}
11.	$a_2,$	$a_4,$	$a_5,$	$a_6,$	$a_7,$	a_{10}
12.	$a_1,$	$a_3,$	$a_5,$	$a_6,$	$a_8,$	a_9

TABLE III

5.	
6.	
7.	$a_9 + a_{11}$
8.	$a_{10} + a_{12}$
9.	$a_7 + a_{12}$
10.	$a_8 + a_{11}$
11.	$a_7 + a_{10}$
12.	$a_8 + a_9$

(Table III) shows that our criterion can no longer be applied.

It is clear that a_5 and a_6 belong to the pseudo core of ϕ because they have no prime consequents.

In brief, we know now that ϕ has a simplest normal equivalent B such that a_5 and a_6 are terms of B , and a_1, a_2, a_3 , and a_4 are not terms of B . Furthermore, the reduced table is so simple, now, that we can tell at a glance that $a_5 + a_6 + a_8 + a_9 + a_{11}$ is a simplest normal equivalent of ϕ .

CONCLUSION

The method that we have just outlined is not an alternative to currently used algorithms for simplifying truth functions. On the contrary, it is meant to be incorporated into these algorithms. After all the unconditional prime implicants of a formula have been detected, and all the prime implicants which are covered by other prime implicants have been deleted, then our algorithm can be applied, and may further reduce the formula. In particular, it may reduce coreless formulas.

Both our routine and the routine to determine the core of a formula have the same object: once they have been

applied, we are forearmed with the knowledge that our formula has a simplest normal equivalent which *includes* a known set of prime implicants, and *excludes* another known set of prime implicants.

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Previous Investigation of Realization of an Arbitrary Switching Function with a Network of Threshold and Parity Elements

Abstract—Attention is called to previous research on realization of an arbitrary switching function by a network of threshold gates (or threshold elements) and modulo-2 gates (or parity elements), establishment of greatest lower bounds on the number of gates needed, artifices that lead to further network reduction in special cases, and systematic minimization of the number of modulo-2 gates required.

Index Terms—Arbitrary switching function, linearly separable function, matrix partitioning, minimal network, modulo-2 gate, optimal realization, parity element, threshold element, threshold gate, Walsh matrix, Walsh series.

Ostapko and Yau recently published a short note on the realization of an arbitrary switching function by a two-level network of threshold and parity elements [1]. This author and others also investigated the realization of an arbitrary switching function by a more general network of these same elements, but followed a very different approach and sought a different form of optimization [2, pp. B-8-B-34].

It is now well known that any m -ary function (not merely binary functions) of n binary variables and 2^n arguments can be expanded into a finite Walsh series. The search for linearly separable threshold functions is simply a search for switching functions that can be represented by the constant and linear terms of their Walsh expansion. In this sense [1, Sect. III] can be viewed as a redevelopment of the widely known Walsh matrix and its binary equivalent [3], [4].

In view of the great decrease in the fraction of switching functions that are linearly separable as n is increased, the author, like Ostapko and Yau, became aware that the search for switching functions realizable by threshold devices alone promised very limited rewards, and that a more fruitful search might be one for optimal means of realizing any switching function by a combination of threshold gates (corresponding to the threshold elements of [1]) for the constant and linear terms in the Walsh expansion, and modulo-2 gates (corresponding to the parity elements of [1]) for the nonlinear terms.

Instead of endeavoring to optimize the realization by programming methods, the author applied the well-known method of partitioning matrices to the recursive Walsh matrix derived by Lechner [3], [4] and showed that any switching function could be realized by a three-level net-

work, the first consisting of at most $2^{n-p} - (n-p) - 1$ modulo-2 gates (where p is the number of partitionings), the second consisting of at most 2^p threshold gates, and the third consisting of a single conventional OR gate [2]. The OR gate is, of course, a special form of threshold element. If there is only one threshold element in the second level, the OR gate is superfluous, and the network becomes identical to that in [1, Fig. 1].

Each partitioning eliminates one variable from the modulo-2 sums in the Walsh expansion and, of course, reduces the number of modulo-2 gates needed. The variables eliminated are then used as control signals to cause each threshold gate to be responsive to a particular subset of 2^{n-p} arguments and inhibited for the remaining $2^n - 2^{n-p}$ arguments.

Assuming each modulo-2 sum in the Walsh expansion to require a separate modulo-2 gate, and assuming all gates of all three types (modulo-2, threshold, and conventional OR) to be of equal cost, regardless of number of inputs and values of weights, the author showed that the total number of gates can be optimized with respect to the number of partitionings. This result has been tabulated for $1 \leq n \leq 12$ and $0 \leq p \leq 11$ [2, p. B-30]. Although this optimization does not necessarily yield a minimal network with respect to number of gates, it does furnish a greatest lower bound for further investigation. It is perfectly general and applicable to totally symmetrical functions as well as less complicated unate and pure threshold functions.

A considerable portion of [2] is devoted to artifices that can be used in various special cases to cause some of the coefficients of the Walsh expansion to vanish identically (by forming a permissible m -ary function from a given binary switching function and its duals with respect to its variables), to reduce the order of certain modulo-2 sums in the Walsh expansion, and to combine certain modulo-2 sums in the expansion to form "nearly constant" functions (whose values are identical for all arguments except 0, 0, ..., 0 or 1, 1, ..., 1 or both, which may be DON'T CARE conditions). Where applicable, the first of these techniques yields smaller greatest lower bounds (hence still more economical networks), which have also been tabulated for the same values of n and p [2, pp. B-31-B-32].

The author's colleague, M. Tannenbaum, showed how the number of modulo-2 sums, hence the number of gates, can always be minimized by a systematic search for a minimal set of acceptable covers [2, pp. B-34-B-66].

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